

Category	Factor	What to Measure	Best View	Injury Risk
KNEE MECHANICS				
Knee Mechanics	Knee Valgus (Caving)	Inward collapse of knees vs. toe line during ascent/descent; flag >10–15° deviation	Frontal	HIGH – ACL/MCL, meniscus
	Knee Tracking	Knees tracking over 2nd–3rd toe throughout full ROM	Frontal	MOD – Patellar tendinitis
	Knee Travel	Forward knee travel past toes; excessive = ankle mobility compensation	Sagittal	LOW – Patellar tendon stress
SQUAT DEPTH & ROM				
Depth	Squat Depth	Hip crease position relative to knee top; measure hip angle at bottom	Sagittal	MOD – Incomplete ROM or lumbar flexion
	Depth Consistency	Rep-to-rep variance in bottom position depth	Sagittal	LOW – Fatigue indicator
SPINAL MECHANICS				
Spine	Butt Wink	Posterior pelvic tilt at bottom; lumbar curvature change from standing baseline >10–15°	Sagittal	HIGH – Disc herniation, nerve impingement
	Thoracic Rounding	Upper back collapse; angle of T-spine relative to vertical	Sagittal	HIGH – Thoracic disc, compensatory hyperextension
	Forward Lean	Torso angle vs. vertical; excessive lean shifts load to lower back	Sagittal	HIGH – Lower back strain, good-morning pattern
	Spinal Neutrality	Maintenance of natural lordotic curve; any hyper-extension or flexion	Sagittal	HIGH – Spondylolisthesis, facet irritation
BAR PATH & POSITION				
Bar Path	Bar Path Verticality	Horizontal deviation of bar from vertical line over midfoot; flag >2–3 in.	Sagittal	MOD – Imbalanced loading
	Bar Path Symmetry	Left-right bar tilt; indicates unilateral weakness	Frontal / Rear	MOD – Asymmetric spinal loading
	Bar Stability	Bar shifting or rolling on back relative to shoulder landmarks	Sagittal / Rear	HIGH – Loss of control, cervical strain
HIP MECHANICS				
Hips	Hip Shift	Lateral hip shift during ascent; >1–2 in. persistent deviation	Frontal	HIGH – SI joint, unilateral overload
	Hinge Initiation	Descent should start with hip hinge before knee bend	Sagittal	MOD – Anterior knee loading

	Hip-Knee-Ankle Stack	Vertical joint alignment in frontal plane throughout rep	Frontal	MOD – Multi-joint stress
TEMPO & CONTROL				
Tempo	Eccentric Tempo	Descent time; <1s at moderate+ load = dive-bombing	Sagittal	HIGH – Rebound injury, loss of control
	Reversal Control	Smoothness at bottom; hard bounce vs. controlled turnaround	Sagittal	MOD – Patellar/Achilles tendon stress
	Concentric Smoothness	Jerky vs. smooth ascent; sticking points indicate weakness	Sagittal	LOW – Stalling under load
	Rep-to-Rep Consistency	Tempo variance across set; degradation signals fatigue	Sagittal	MOD – Fatigue-driven breakdown
FOOT & ANKLE				
Foot/Ankle	Foot Pressure	Weight distribution across tripod foot (heel, ball, toes)	Sagittal / Frontal	MOD – Forward/backward pitch
	Heel Rise	Heels lifting off floor; indicates ankle dorsiflexion limitation	Sagittal	MOD – Forward pitch, lumbar compensation
	Foot Rotation	Feet turning in/out during rep vs. starting position	Overhead / Frontal	MOD – Ankle/knee torsion
	Stance Width Shift	Feet drifting wider or narrower mid-rep	Frontal	LOW – Base instability
UPPER BODY POSITION				
Upper Body	Head/Neck Position	Cervical alignment; neutral gaze vs. excessive extension or flexion	Sagittal	MOD – Cervical strain, balance disruption
	Elbow Position	Elbows pointing down (good) vs. flared back; lat engagement cue	Rear	LOW – Shoulder impingement, bar instability
	Wrist Alignment	Neutral wrist vs. excessive extension; wrists support, not bear load	Rear / Sagittal	LOW – Wrist tendinitis
	Shoulder Retraction	Scapulae retracted and depressed to create stable bar shelf	Rear	LOW – Bar instability, upper back rounding
BREATHING & BRACING				
Bracing	Brace Quality	360° trunk expansion vs. chest-only breathing before descent	Sagittal / Frontal	HIGH – Spinal instability without IAP
	Brace Timing	Breath taken and held before descent begins	Sagittal	HIGH – Unprotected spine at bottom
	Brace Maintenance	IAP held through sticking point; visible collapse = loss	Sagittal	HIGH – Sudden spinal deloading

BILATERAL SYMMETRY				
Symmetry	Left/Right Symmetry	Compare all landmarks left vs. right; one side rising faster or dipping	Frontal	HIGH – Cumulative asymmetric loading
	Hip Rise Rate	Both hips rising at same rate; if hips outpace chest = good-morning	Sagittal / Frontal	HIGH – Extreme lower back stress